INCH-POUND

MIL-DTL-17/235
21 June 2017

DETAIL SPECIFICATION SHEET

CABLES, RADIO FREQUENCY, FLEXIBLE, COAXIAL, 75 OHMS, LOW SMOKE, ENHANCED INSULATION RESISTANCE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-17.

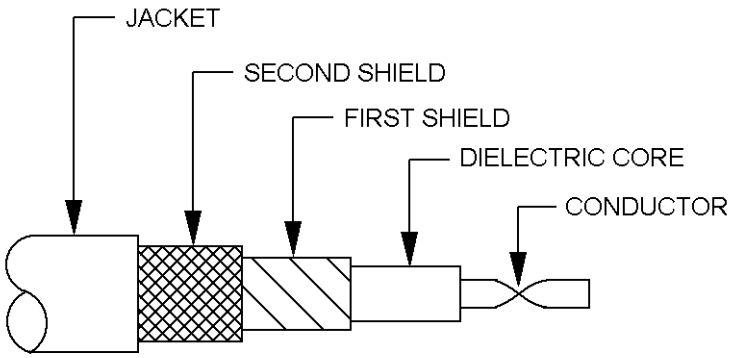


FIGURE 1. General configuration.

TABLE I. Description.

Components	Construction details
Inner conductor	24 AWG 19x36 silver copper plated alloy Overall diameter: .024 ± .001 nominal
Dielectric Core	Expanded PTFE/PTFE tape composite Diameter: .100 inch ± .004.
First shield	Silver plated copper braid having 95% minimum coverage. .108 inch nominal diameter.
Second shield	38 AWG silver plated copper braid having 95% minimum coverage. .126 inch nominal diameter.
Jacket	Extruded fluorinated ethylene propylene (FEP) Diameter: .150 inch ± .005.

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ENGINEERING INFORMATION:

Continuous working voltage: 750 V rms, maximum.

Operating frequency: 3 GHz, maximum.

Velocity of propagation: 79 percent, nominal.

Operating temperature range: -55° to +200°C.

Inner conductor properties:

DC resistance, maximum, at 20°C: 2.74 ohms per 100 feet.

Elongation: 6 percent, minimum.

Tensile strength: 50 $klb_f/inch^2$, minimum.

REQUIREMENTS:

Dimensions, configuration and description: See figure 1.

Environmental and mechanical:

Visual and mechanical examination: Applicable.

Adhesion of conductors:

Inner conductor to core: 1 pounds, minimum; 10 pounds maximum.

Aging stability: +98° ±2°C.

Stress crack resistance: 230° C.

Outer conductor integrity: Not applicable.

Cold bend: -55°C ±2°C.

Dimensional stability: +200°C ±5C.

Inner conductor from core: 0.125 inch, maximum.

Inner conductor from jacket: .187 inch, maximum.

Contamination: Not applicable.

Flame propagation: Applicable.

Weight: 0.235 Lbs/ft, maximum.

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Electrical:

Spark test: 1,500 Vrms, +25 percent, -0 percent.

Voltage withstanding: 750 Vrms, minimum.

Corona extinction voltage: 1,200 V rms, minimum.

Characteristic impedance: 75 ±10 ohms.

Attenuation: 100MHz – 4.3dB/100ft
400MHz – 8.7dB/100ft
500MHz – 9.8dB/100ft
1GHz – 14.2dB/100ft
2GHz – 21.0dB/100ft

Structural return loss: Not applicable.

Capacitance: 17.5pF per foot, maximum.

Capacitance unbalance: Not applicable.

Transmission unbalance: Not applicable.

Time delay: 1.30Ns/ft (nominal).

Part or Identifying Number (PIN): M17/235-00001

Referenced documents. This document references MIL-DTL-17.

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CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force – 85
DLA - CC

Review activities:
Army - AR, AT, CR4, MI
Navy - AS, MC, OS, SH
Air Force - 19, 99

Preparing activity:
DLA - CC

(Project: 6145-2017-024)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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